

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			[9 + 14 =] 23 [m] (1)		1		1		
	(b)			As the speed increases, the thinking distance increases (1) {at a constant rate / linearly / proportionately} (1) Or Thinking distance is proportional to speed (2) Or If the speed doubles, the thinking distance doubles OWTTE (2) Or For every 10 mph increase in speed, thinking distance increases by 3 m (2) Treat as neutral any reference to braking distance		2		2	2	

Question				Marking details	Marks available					
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	(c)			Indicative content: Thinking distance is the distance travelled in the driver’s reaction time. Braking distance is the distance travelled while braking. Factors: Thinking distance is affected by alcohol, drugs, tiredness, use of mobile phone and the speed of the car. Braking distance is affected by the condition of the road, the condition of the brakes and tyres and the speed of the car. 5 – 6 marks Correctly describes what is meant by both distances and gives some correct examples of the factors affecting each of them. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 marks Correctly describes what is meant by one of the distances and gives correct examples of the factors affecting that distance OR gives a partial description of both with some factors. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1–2 marks Correctly describes what is meant by one of the distances OR gives correct examples of the factors affecting that distance OR gives some factors that can affect either distance. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	6			6		
				Question 6 total	6	3	0	9	2	0